Online Book Store Analytics Using SQL

An End-to-End SQL Data Analytics Portfolio Project

Domain: E-Commerce | Retail | Data Analytics

Database: Online Book Store Database

Tools: MySQL · SQL · MySQL Workbench

Type: Self-Learning SQL Portfolio Project

📌 Project Overview & Business Problem

The management of an online bookstore collects thousands of transaction records every year. Although sales data is continuously generated, the company struggles to convert this raw data into meaningful business insights.

The management team wanted answers to several critical business questions:

- Which books generate the highest revenue?
- Which genres perform best?
- Who are the most valuable customers?
- Which books are not selling?
- Are there inventory shortages?
- Which months generate the highest revenue?
- How can sales and profitability be improved?

> **Without proper analysis**, management was making business decisions based on assumptions instead of data.

🚨 Business Challenges

📚 Product Challenges

- Difficulty identifying best-selling books
- No visibility into underperforming books
- Poor inventory planning

👥 Customer Challenges

- Unable to identify valuable customers
- No customer segmentation
- Difficulty creating loyalty programs

💰 Revenue Challenges

- No understanding of monthly sales trends
- Unable to identify seasonal demand
- Lack of revenue analysis by author or genre

📦 Inventory Challenges

- Books going out of stock
- Unsold inventory increasing storage costs
- No inventory monitoring system

📈 Business Challenges

- Large amount of raw data
- Manual reporting
- Slow decision-making
- No centralized business insights

🎯 Project Objectives

The objective of this project was to transform raw bookstore data into actionable business insights using SQL. The project focused on:

- Designing a relational database
- Performing exploratory data analysis
- Solving business problems using SQL
- Creating reusable SQL queries
- Generating management-ready business reports
- Supporting data-driven decision-making

📂 Dataset & Database Design

The project consists of three relational tables, each capturing a distinct part of the bookstore's operations.

Table	Fields
📚 **Books**	Book ID, Title, Author, Genre, Price, Published Year, Stock
👤 **Customers**	Customer ID, Name, Email, Phone, City, Country
🛒 **Orders**	Order ID, Customer ID, Book ID, Quantity, Total Amount, Order Date

🏗 Database Design

The database follows a normalized relational structure:

```
   Customers
       |
   | Customer_ID
       v
     Orders
       ^
     | Book_ID
       |
      Books
```

Relationships:

- One Customer → Many Orders
- One Book → Many Orders

🛠 SQL Workflow & Project Phases

The project followed a complete, end-to-end SQL analytics workflow:

Business Problem → Database Design → Import Dataset → Data Cleaning → Exploratory Data Analysis
↓
Business Analysis → JOIN Operations → Advanced SQL → Views → Stored Procedures → Indexes → Final Business Insights

📖 Project Phases

	Focus Area	Details
1	Business Understanding	Defining the business problem and key questions
2	Dataset Understanding	Exploring the structure of the raw data
3	Database Design	Building the normalized relational schema
4	Data Import	Loading data into MySQL
5	Exploratory Data Analysis	COUNT, AVG, SUM, MIN, MAX, DISTINCT
6	Business Analysis	Highest revenue books, genre analysis, customer analysis, sales trends
7	SQL JOINs	INNER JOIN, LEFT JOIN, RIGHT JOIN, multiple table JOINs
8	Advanced SQL	Subqueries, CTEs, window functions, RANK(), DENSE_RANK(), ROW_NUMBER(), running totals
9	Database Optimization	Views, stored procedures, indexes, EXPLAIN, query optimization
10	Business Insights Report	Management-ready business report based on SQL analysis

🔍 SQL Concepts Used

Basic SQL

SELECT **WHERE** **ORDER BY** **GROUP BY** **HAVING** **LIMIT**

Aggregate Functions

COUNT() **SUM()** **AVG()** **MIN()** **MAX()**

Joins

INNER JOIN **LEFT JOIN**

Advanced SQL

Subqueries **CTE** **Window Functions**

Database Objects

Views **Stored Procedures** **Indexes**

📊 Business Insights

$75,628.66	500
TOTAL REVENUE GENERATED	CUSTOMER ORDERS

Product Analysis

- Top-selling books
- Lowest-selling books
- Revenue by author
- Revenue by genre

Customer Analysis

- Highest spending customers
- Customers with multiple orders
- Customers without purchases

Sales Analysis

Monthly revenue trends revealed:

- Peak sales months
- Low-performing months
- Seasonal variations

Inventory Analysis

The SQL analysis identified:

- Out-of-stock books
- Low-stock books
- Unsold books

💡 Recommendations & Project Outcomes

Business Recommendations

📦 Inventory

- Restock high-demand books
- Monitor inventory regularly
- Reduce stock of slow-moving books

📣 Marketing

- Promote unsold books
- Create genre-based marketing campaigns
- Offer seasonal discounts

👥 Customer Strategy

- Launch loyalty programs
- Reward top customers
- Re-engage inactive customers

📈 Sales

- Increase focus on high-performing genres
- Monitor monthly sales trends
- Expand popular book categories

📈 Project Outcomes

The SQL analysis enabled the business to:

- Understand customer purchasing behavior
- Identify best-selling products
- Monitor inventory effectively
- Improve revenue visibility
- Support data-driven decision-making
- Reduce manual reporting effort

🎓 Skills Demonstrated & Key Learnings

SQL Skills

- Database Design
- Data Analysis
- Aggregations
- Joins
- Subqueries
- CTEs
- Window Functions
- Views
- Stored Procedures
- Indexes

Analytical Skills

- Business Analysis
- KPI Development
- Revenue Analysis
- Customer Segmentation
- Inventory Analysis
- Trend Analysis
- Problem Solving

🏆 Key Learnings

This project strengthened practical knowledge in:

- Relational database design
- Writing optimized SQL queries
- Solving real-world business problems
- Extracting actionable insights from raw data
- Converting business questions into SQL queries
- Presenting analytical findings in a structured report

🚀 Conclusion

This project demonstrates a complete SQL analytics workflow, from understanding a business problem to delivering actionable insights. By analyzing sales, customers, products, and inventory through SQL, the project shows how structured query techniques can support strategic decision-making in an online retail business.

> The outcome is a reusable, business-oriented SQL portfolio project that highlights database design, analytical thinking, query optimization, and insight generation — skills that are directly applicable to entry-level Data Analyst and SQL Developer roles.